

# AI-Powered Discovery — Cheat Sheet

Everything from Module 3, one page. The throughline: **show me the source**.

## 1 What Discovery Answers

### Is the problem real?

Or just loud in the one meeting it came up in.

### Who has it, and how many?

A market, or an anecdote.

### What already exists?

Someone met this problem before you did.

**Why you care:** if a piece of research moves none of the three, it isn't research. It's reading.

## 2 The One Rule

**In discovery, an answer without a source is a liability.**

**Grounded, not generated.** In Module 2 the model wrote from its own head and you were the source. Here it makes claims about the real world.

### RAG you can see

Both tools fetch real sources **first**, then answer, then hand you the citation. Retrieval before generation is Module 1's answer to hallucination.

**Why you care:** a confident unsourced number is how you talk yourself into a quarter of wrong work.

## 3 The Failure Mode Flipped

Old risk: doing ~~too little~~ research, because it was slow.

New risk: a confident, completely wrong picture of your market in ninety seconds, believed because it reads well.

**Lazy → gullible.**

**Why you care:** gathering is now nearly free, so it is no longer where your value sits. **Judgement is the job.** The tools read; you decide what's true.

## 4 Three Sources, Three Tools

|                   |                                              |
|-------------------|----------------------------------------------|
| <b>Market</b>     | outside your walls · <b>Perplexity</b> · 3.2 |
| <b>Your docs</b>  | what you own · <b>Gemini Notebook</b> · 3.3  |
| <b>User voice</b> | transcripts · <b>your copilot</b> · 3.4      |

**Why you care:** a brief built on one source type is an opinion. Three that agree is a finding.

## 12 What You Walk Away With

A **one-page research brief** on a real problem, built from all three source types, every claim clickable back to where it came from.

Problem · evidence · who and how many · white space · recommendation · **what you're still unsure about**.

**Why you care:** pair it with your Module 2 PRD. The brief is the evidence for the question; the PRD is the answer. Together, the front half of your capstone.

## 5 Perplexity: Cited Research

### The three-second habit

**Click the citation.** Before any fact goes in a doc with your name on it, open the source.

**Would I be comfortable if my VP clicked this same link?**

### Depth ladder

|                             |                           |                           |
|-----------------------------|---------------------------|---------------------------|
| <b>QUICK</b><br>a fact, now | <b>PRO</b><br>a landscape | <b>DEEP</b><br>a teardown |
|-----------------------------|---------------------------|---------------------------|

**Names and limits move.** The workflow is durable; the buttons are not. Check the monthly update.

**Why you care:** the citation is the product, not the paragraph. Most people never click, and clicking is the difference between research and vibes.

## 6 Grading a Source

- ✗ **Competitor's own page** proves their positioning, not that it works.
- ✗ **Content-farm blog** or a number with no origin: that's a lead, not a fact.
- ✗ **The year chain:** a 2026 article citing a 2019 estimate is a 2019 number.
- ✓ **Primary and dated:** filings, regulators, the original study, the actual page.
- ✓ **Complaints:** one-star reviews and support threads are where white space hides.

**Why you care:** the summary is not the page. If the summary and the source disagree, the source wins.

## 7 Gemini Notebook: Your Own Corpus

Renamed from NotebookLM. Same product, same workflow. Locked to the sources you load and cited to the passage. It searches nothing else, so it can't wander onto the web. **The cage is the feature.**

### One question, one notebook

Not one giant archive. Load PRDs, memos, tickets, sales notes, transcripts.

### The three asks

1. Institutional memory: **what did we learn about this and forget?**
2. A grounded briefing where every claim links back.
3. **Where do these sources disagree?**

**Why you care:** the gap between last year's memo and this year's tickets is not noise. That gap is the insight you walk into the room with.

## 8 Grounded ≠ Right

A citation proves the sentence **is in your document**. It does not prove your document was **right**.

Load a stale memo or a contradictory ticket dump and it reports that mess back faithfully, with a link.

**Why you care:** assume nothing in your own drive is coherent until you've checked. Your archive is an input, not an authority.

## 9 Interviews Into Themes

**Thematic analysis, four moves:** read → code → cluster → name. AI is genuinely good at the middle two. The first and last stay yours.

Tireless at 4pm, reads the twelfth transcript like the first, spots an echo across weeks. It does not know what matters to your roadmap.

**Why you care:** you own the reading and the naming, because the name is what you say out loud in front of your VP.

## ! Six Ways Synthesis Lies

|                   |                                         |
|-------------------|-----------------------------------------|
| <b>Invents</b>    | plausible themes not in the data        |
| <b>Buries</b>     | the lone dissenter, averaged away       |
| <b>Flattens</b>   | "love it, but I'd never pay" → positive |
| <b>Obeys</b>      | your framing: ask for pain, get pain    |
| <b>Skims</b>      | the middle; ends and vivid language win |
| <b>Fabricates</b> | quotes: in character, never said        |

**Why you care:** the insight is usually in the **but**, and the dissenter is often your sharpest signal. Both are what averaging destroys.

## 10 The Two Prompts That Fix It

**No quote, no theme. Give me the exact words and which participant said them.**

**What in these transcripts contradicts each theme you just gave me?**

An invented theme can't produce a real quote, so it collapses when you ask. Then **spot-check one quote in seven** against the transcript: really said, really means that.

Then triage, like Module 2's edge cases: ~~kill the thin theme~~ ("users want a less noisy experience" is true of every product ever built).

**Why you care:** a paraphrase inside quotation marks reaches slide 4, and eight weeks later somebody asks which interview it came from.

## 11 Before You Paste

Transcripts are real people. **Strip names, employers and contact details** before any transcript goes into a tool.

Module 2's tier test still applies: consumer tier may train on it, so real customer words belong in a company-sanctioned account.

**Why you care:** it's your name on the research and on the privacy incident.

## PHRASES THAT SHOULD MAKE YOU PAUSE

### "The market is worth \$4B"

No link, no year, no method. Ask which source, from when.

### "Users said..."

Which user, in which interview, in what words? Ask for the verbatim.

### "Sentiment was positive"

Averaging. Ask what the strongest objection was, and who made it.

### "Studies show" / "Sources indicate"

A citation shape with no citation. Open it or drop it.

### A quote that reads too clean

Real speech has "um" and false starts. Tidy quotes are often paraphrase.